

ES — Elements from the Sea

OCR B (Salters) H433 storyline **ES**. Chemistry from sea water: halogen chemistry, redox and electrolysis, the introduction of equilibrium, and atom economy.

1. Oxidation States and Redox Reactions

Oxidation States (Oxidation Numbers)

Oxidation state: A number assigned to an atom in a compound that represents the number of electrons it has lost, gained, or shared unequally compared to the uncombined element. It is a bookkeeping device for tracking electron transfer.

Rules for assigning oxidation states:

- Uncombined elements have an oxidation state of 0.
- In simple ions, the oxidation state equals the charge (e.g. $\text{Na}^+ = +1$, $\text{Cl}^- = -1$).
- Oxygen is -2 (except in peroxides where it is -1 , and in OF_2 where it is $+2$).
- Hydrogen is $+1$ (except in metal hydrides where it is -1).
- Fluorine is always -1 .
- The sum of oxidation states in a neutral compound is 0; in a polyatomic ion it equals the ionic charge.

Redox Definitions

Oxidation: Loss of electrons / increase in oxidation state.

Reduction: Gain of electrons / decrease in oxidation state.

Reducing agent: A substance that donates electrons (is itself oxidised).

Oxidising agent: A substance that accepts electrons (is itself reduced).

Remember: OIL RIG — Oxidation Is Loss, Reduction Is Gain (of electrons).

Half-Equations

A redox reaction can be split into two half-equations: one showing oxidation and one showing reduction. To write and balance half-equations:

1. Write the species being oxidised or reduced on each side.
2. Balance atoms other than O and H.
3. Balance oxygen by adding H_2O .
4. Balance hydrogen by adding H^+ .
5. Balance charge by adding electrons (e^-).

To combine half-equations, multiply each so the electrons cancel, then add the two equations together and simplify.

Exam Tip: When balancing redox equations, always check that atoms and charges balance on both sides. A common mistake is forgetting to balance hydrogen atoms after adding water molecules. Also check that the electrons cancel completely when combining half-equations.

2. Dynamic Equilibrium

Dynamic equilibrium: A state reached in a closed system when the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant but are not necessarily equal. Both forward and reverse reactions continue to occur at the molecular level.

Conditions for dynamic equilibrium:

- The system must be closed (no substances can enter or leave).
- The forward and reverse rates must be equal.

- The concentrations of all species remain constant over time.
- The reaction must be reversible.

Le Chatelier's Principle

Le Chatelier's principle: If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium will shift in the direction that tends to oppose the change, so as to partially restore the original conditions.

Change	Effect on equilibrium position	Effect on K
Increase concentration of a reactant	Shifts to the right (towards products)	No change
Decrease concentration of a product	Shifts to the right	No change
Increase pressure	Shifts towards the side with fewer moles of gas	No change
Increase temperature	Shifts in the endothermic direction	Changes (increases if forward reaction is endothermic)
Add a catalyst	No change (speeds up both forward and reverse equally)	No change

Exam Tip: A catalyst does not change the position of equilibrium or the value of K. It only increases the rate at which equilibrium is reached. Temperature is the only factor that changes the value of the equilibrium constant.

Le Chatelier's Principle — Equilibrium Shifts

← **Reactants****Equilibrium Position****Products** →Increase $[\text{N}_2]$

Shifts right

Increase pressure

Shifts right
(fewer gas moles)

Increase temperature

Shifts left (endo)

Add catalyst

NO SHIFT

Faster only

Concentration & pressure changes: K unchanged, position shifts to oppose change
Temperature changes: K changes, shifts in the endothermic direction